CSA0714 — Computer Networks

Common Assignment Report

Evaluating a Smart Healthcare Network Through Protocol Analysis, Topology Selection, and Simulation-Based Performance Assessment

Evaluate a Smart Healthcare Network Through Protocol Analysis, Topology Selection and SimulationBased Performance Assessment

Submitted by (Team):
RITESH.Y — 192512100
KIRAN.M — 192512401
CHIRANJIBI PRADHAN — 192511475

This copy: Individual Report of CHIRANJIBI PRADHAN (192511475)

Course Outcomes Mapped: CO3, CO4, CO5
SDG Mapping: SDG 3 — Good Health and Well-Being
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1. Problem Statement and Problem Formulation

Modern multi-floor hospitals depend on a continuously available, low-latency communication network to support patient monitoring, electronic medical records (EMR), laboratory information systems, pharmacy services, and emergency communication. A failure or slowdown in this network can directly delay clinical decisions, so the underlying network architecture must be engineered — not assumed — for reliability, scalability, and predictable performance under load.

This assignment requires the design, simulation, and evaluation of a communication network for a hypothetical multi-floor hospital consisting of at least four interconnected network zones (e.g., Patient Wards, ICU/Monitoring, Diagnostics & Laboratory, Administration/Pharmacy) and a minimum of 40 end devices spanning computers, patient-monitoring terminals, servers, and other connected healthcare equipment. Four candidate topologies — star, mesh, bus, and hybrid — must be constructed and compared, one architecture selected and implemented in a network simulator, and its behaviour evaluated under TCP, UDP, HTTP, DNS, FTP, and ICMP traffic across at least three experimental scenarios (normal operation, peak patient-monitoring load, and network disruption/congestion).

The core problem this report addresses is therefore twofold: (a) which physical/logical topology best balances reliability, scalability, cost, and resource utilization for a healthcare environment of this scale, and (b) how core network protocols behave — and degrade — under realistic healthcare traffic conditions, so that a defensible, evidence-based architecture recommendation can be made.

2. Objectives and Expected Outcomes

- Identify the communication requirements of a multi-floor hospital across at least four network zones and 40+ end devices.
- Design and construct four candidate topologies — star, mesh, bus, and hybrid — using appropriate switches, routers, transmission media, and cabling standards.
- Select and implement the most suitable architecture in a network simulation platform (Packet Tracer / GNS3 / EVE-NG).
- Generate and capture TCP, UDP, HTTP, DNS, FTP, and ICMP traffic representative of healthcare applications under normal, peak-load, and disrupted-network scenarios.
- Measure at least five performance indicators (e.g., response time, throughput, packet delivery ratio, packet loss, jitter, end-to-end delay, bandwidth utilization) across repeated trials.
- Interpret the results to identify bottlenecks and congestion points, and recommend the most suitable topology and protocol configuration with quantitative justification.

Expected outcome: a validated, simulation-backed recommendation for the hospital's network architecture, supported by reproducible test data, comparative analysis, and an engineering justification that explicitly ties design choices to measured evidence.

3. Requirements, Constraints and Assumptions

3.1 Functional Requirements

- Continuous, low-latency delivery of patient-monitoring data (vital signs) from bedside devices to the central monitoring station.
- Reliable transfer and storage of electronic medical records (EMR) between wards, labs, and the records server.
- Timely laboratory result delivery between diagnostic equipment and physician workstations.
- Pharmacy inventory and prescription communication between pharmacy systems and ward terminals.
- Priority handling of emergency/code-blue communication traffic.

3.2 Non-Functional Requirements

- High availability and fault tolerance — no single link/device failure should isolate a critical zone.
- Scalability to accommodate additional floors, devices, or services without a redesign.
- Bounded latency and jitter for real-time monitoring traffic.
- Efficient bandwidth utilization under peak concurrent load.

3.3 Constraints

- Minimum of 4 interconnected network zones and 40 end devices (as specified in the assignment brief).
- Must use standardized cabling (e.g., Cat6 UTP, fiber backbone where justified) and campus-appropriate devices (Layer 2/3 switches, routers, access points).
- Simulation constrained to the features/device models supported by the chosen simulator.

3.4 Assumptions

- The hospital building has 4 floors, each hosting one primary network zone, connected via a backbone/core switch or router.
- Background office traffic (email, browsing) coexists with clinical traffic and is modelled to create realistic congestion.
- Simulated link speeds and device counts are scaled to what the chosen simulator can reliably execute while remaining representative.

> *PLACEHOLDER — replace with your team's actual work: state your team's actual assumed floor plan, zone names, and device counts here if they differ from the above.*

4. Application of Relevant Course Knowledge / Concepts

Course Concept	Application in this Assignment
Network topologies (star, mesh, bus, hybrid)	Compared structurally and by device/link count for the hospital layout
OSI / TCP-IP layering	Used to reason about where each protocol (TCP, UDP, HTTP, DNS, FTP, ICMP) operates and interacts with the topology
Transport layer protocols (TCP vs UDP)	TCP used for reliable EMR/file transfer; UDP modelled for time-sensitive monitoring data
Application layer protocols (HTTP, DNS, FTP)	HTTP for web-based hospital portals, DNS for name resolution of internal servers, FTP for bulk record/image transfer
Network layer / ICMP	Used for reachability testing (ping) and fault diagnosis during the

	disruption scenario
Structured cabling & transmission media	Cat6 UTP for horizontal runs, fiber for inter-floor backbone, justified by distance/bandwidth needs
Performance metrics (throughput, delay, jitter, loss, PDR)	Directly measured in the simulator to compare topologies and protocol behaviour
Network devices (switches, routers, access points)	Selected per zone based on required port density, Layer 2 vs Layer 3 functionality, and redundancy needs

5. Design / Proposed Solution / Methodology

Four candidate topologies were designed for the hospital's four network zones (Ward/Patient Monitoring, ICU, Laboratory & Diagnostics, Administration/Pharmacy), each interconnected through a core layer. Logical and physical diagrams for each topology (devices, links, media, and cabling standards) are provided in Appendix A / the accompanying simulation project files.

5.1 Star Topology

Each zone's end devices connect to a dedicated access-layer switch; the four zone switches connect to a central core switch/router.

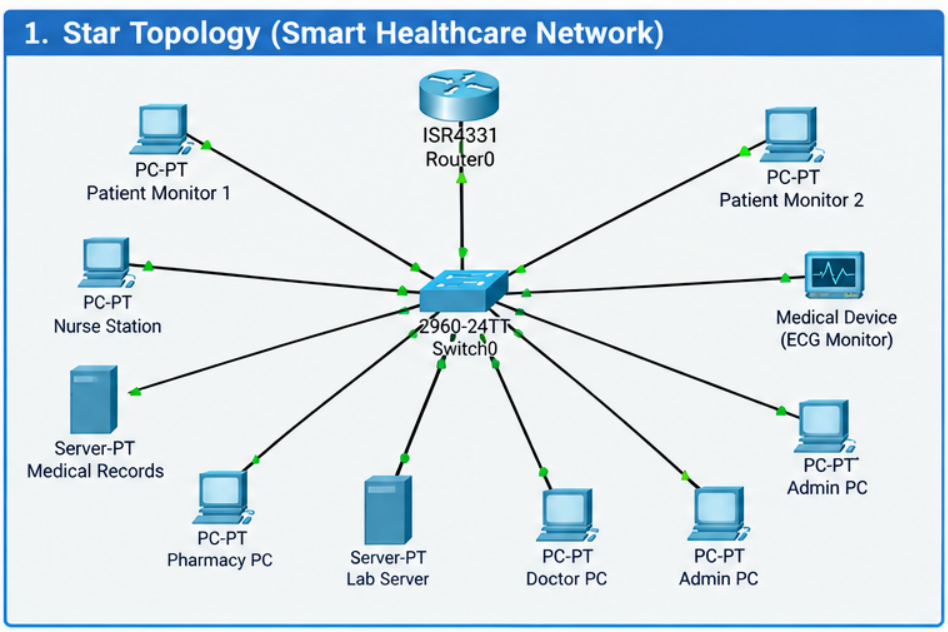

Figure 5.1 — Star topology diagram for the Smart Healthcare Network (team-prepared diagram; replace/supplement with your Packet Tracer implementation screenshot).

Aspect	Description
Devices	1 core switch/router + 4 access switches (one per zone) + 40+ end devices
Media	Cat6 UTP (end devices to access switch); Cat6/fiber (access switch to core)
Advantages	Simple to manage and troubleshoot; single device failure only affects one node/zone

Disadvantages	Core switch is a single point of failure for the whole network

5.2 Mesh Topology

Zone routers/switches are interconnected with redundant links (full or partial mesh) so that multiple paths exist between any two zones.

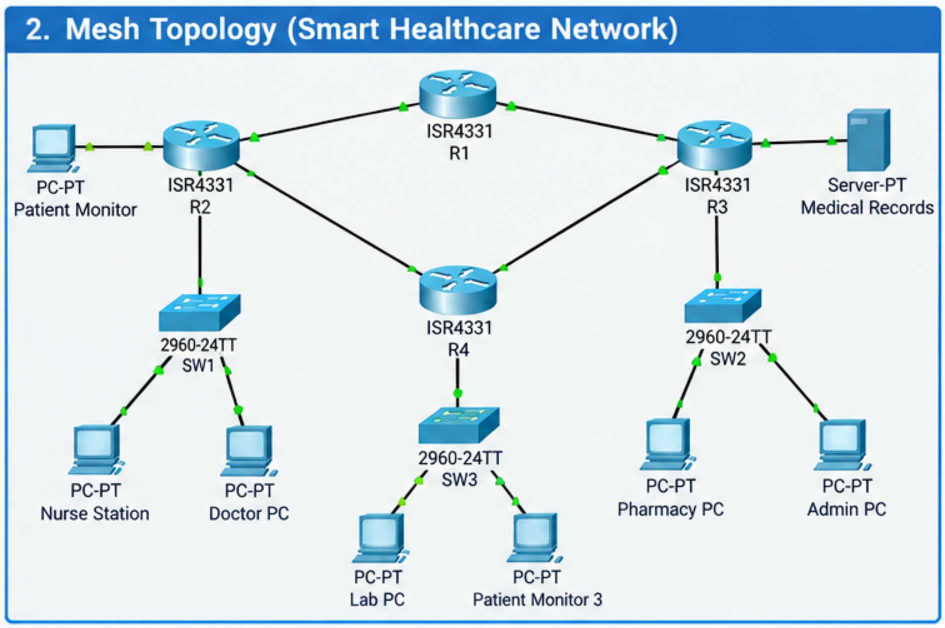

Figure 5.2 — Mesh topology diagram for the Smart Healthcare Network (team-prepared diagram; replace/supplement with your Packet Tracer implementation screenshot).

Aspect	Description
Devices	4 zone routers/L3 switches interconnected (partial mesh recommended at this scale) + access switches + end devices
Media	Fiber or Cat6 for inter-zone links; UTP for end-device access
Advantages	High fault tolerance and redundancy — critical for emergency communication
Disadvantages	Higher cabling/device cost and configuration complexity; harder to scale

5.3 Bus Topology

All devices in a zone are attached to a single shared backbone cable via a terminated trunk (modelled in-simulator using a shared segment/hub where the platform requires it).

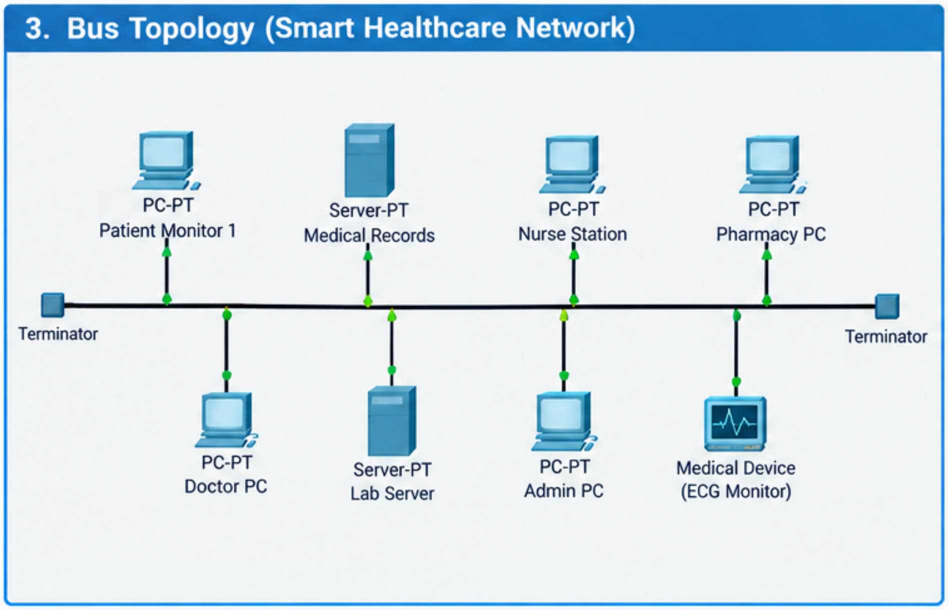

Figure 5.3 — Bus topology diagram for the Smart Healthcare Network (team-prepared diagram; replace/supplement with your Packet Tracer implementation screenshot).

Aspect	Description
Devices	Shared backbone medium per zone; minimal switching hardware
Media	Coaxial/UTP shared segment (legacy reference topology)
Advantages	Low cabling cost, simple to deploy for a small segment
Disadvantages	A single backbone fault disables the entire zone; poor scalability and collision domain issues — not recommended for critical healthcare traffic

5.4 Hybrid Topology (Recommended Candidate)

Combines a redundant mesh/backbone core (for inter-zone and emergency traffic resilience) with star-wired access layers within each zone (for simplicity and ease of device management).

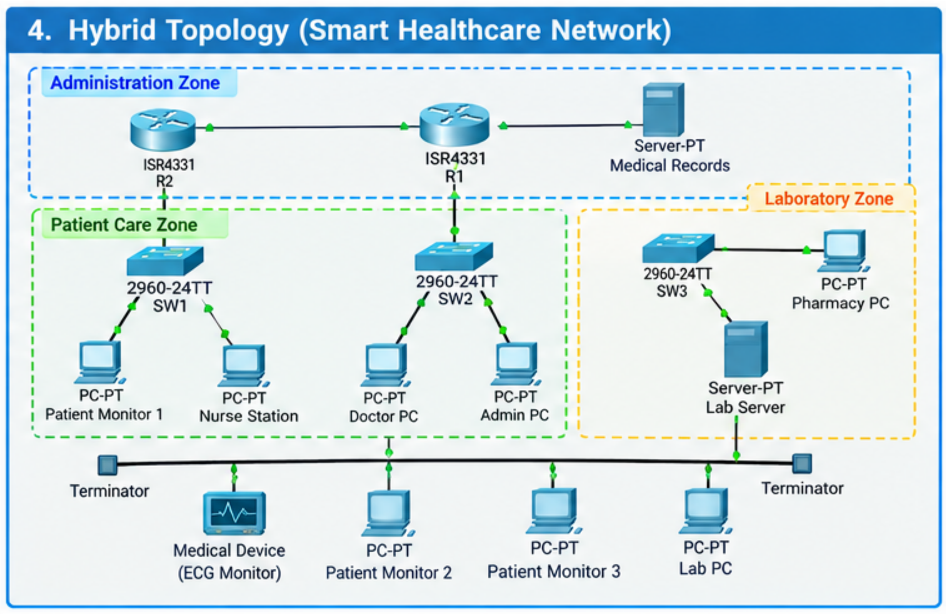

Figure 5.4 — Hybrid topology diagram for the Smart Healthcare Network (team-prepared diagram; replace/supplement with your Packet Tracer implementation screenshot).

Aspect	Description
Devices	Redundant core (2 core switches/routers), 4 zone access switches (star-wired), 40+ end devices
Media	Fiber backbone between core devices; Cat6 UTP for zone access
Advantages	Combines fault tolerance of mesh with manageability of star; scalable per zone
Disadvantages	More complex to design and document than a single pure topology

5.5 Selected Architecture and Justification

Based on the reliability, scalability, and resource-utilization criteria in the assignment brief, the hybrid topology is proposed as the primary implementation candidate: it avoids the single point of failure of the star design and the whole-segment outage risk of bus, while remaining more cost-effective and easier to manage than a full mesh at this scale.

> *PLACEHOLDER — replace with your team's actual work: confirm this selection against your own comparative results in Section 9 — replace with your team's actual justification once simulation data is available.*

6. Experiment and Test Plan

6.1 Test Scenarios

#	Scenario	Description	Purpose
1	Normal Operation	Baseline clinical + admin traffic at typical hourly load	Establish baseline performance for each protocol
2	Peak Patient-Monitoring Traffic	Concurrent UDP vital-sign streams from all ICU/ward monitors + background HTTP/FTP	Assess behaviour under high concurrent real-time load

3	Network Disruption / Congestion	Simulated link failure (mesh/hybrid reroute) or induced congestion (bandwidth saturation)	Evaluate resilience, failover time, and degradation

6.2 Variables and Metrics

- Independent variables: topology type, traffic type (TCP/UDP/HTTP/DNS/FTP/ICMP), scenario (normal/peak/disrupted).
- Dependent variables (performance indicators): response time, throughput, packet delivery ratio, packet loss, jitter, end-to-end delay, bandwidth utilization (select ≥5).

6.3 Repetition Strategy

Each scenario is run a minimum of 3 trials per topology/protocol combination; the mean and standard deviation of each metric are reported to improve reliability of the observations and reduce the effect of simulator randomness.

> *PLACEHOLDER — replace with your team's actual work: record your actual trial count and any deviations from this plan here.*

7. Algorithm / Pseudocode

The following pseudocode outlines the traffic-generation and measurement-collection procedure used across all simulation trials.

```
BEGIN SimulationExperiment
  INPUT: topology_type, scenario_type, trial_count
  FOR each topology IN [Star, Mesh, Bus, Hybrid]:
      BUILD network(topology) WITH zones=4, end_devices>=40
      CONFIGURE devices: IP addressing, DNS server, routing/switching
      FOR each scenario IN [Normal, PeakMonitoring, Disruption]:
          FOR trial = 1 TO trial_count:
              START traffic generators:
                  UDP  -> vital-sign monitoring streams (periodic small packets)
                  TCP  -> EMR/file transfer between ward PC and records server
                  HTTP -> hospital portal requests to web server
                  DNS  -> name resolution queries for internal hostnames
                  FTP  -> bulk lab-image / record transfer
                  ICMP -> periodic reachability (ping) checks
              IF scenario == Disruption:
                  INDUCE link failure OR bandwidth saturation on core link
              CAPTURE packet trace / simulator statistics
              COMPUTE metrics: response_time, throughput, PDR,
                               packet_loss, jitter, end_to_end_delay
              STORE (topology, scenario, trial, metrics) -> results_log
          END FOR
      END FOR
  END FOR
  AGGREGATE results_log BY (topology, scenario, protocol) -> mean, std_dev
  RANK topologies BY weighted criteria (reliability, scalability, performance)
  OUTPUT comparative tables and recommended topology
END SimulationExperiment
```

8. Implementation / Simulation Environment and Tools Used

Item	Details
Simulation Platform	[e.g., Cisco Packet Tracer 8.x / GNS3 / EVE-NG]
Devices Modelled	Core router(s), Layer 2/3 switches, servers (DNS, Web/HTTP, FTP, EMR/database), end-user PCs, monitoring terminals
Addressing Scheme	[Insert VLAN / subnet plan per zone]
Traffic Generation Method	[Simulator traffic generator / custom PDU generation / scripted pings and transfers]
Measurement Method	[Simulator statistics panel, Wireshark capture within simulator, custom logging]

PLACEHOLDER — replace with your team's actual work: fill in this table with your actual tool, addressing plan, and measurement method, and attach the simulator project files as required deliverable #3.

9. Results and Validation

The tables below present the format for reporting measured performance across topologies and scenarios. Replace the illustrative values with your team's actual measured results (mean of ≥3 trials), and include the corresponding charts.

9.1 Results Table — Normal Operation

PLACEHOLDER — replace with your team's actual work: illustrative example values below — replace with your own measured readings (mean of ≥3 trials) from the simulator.

Topology	Protocol	Avg. Response Time (ms)	Throughput (kbps)	Packet Loss (%)	Jitter (ms)	PDR (%)
Star	TCP/HTTP	42	870	0.3	4	99.7
Mesh	TCP/HTTP	38	910	0.1	3	99.9
Bus	TCP/HTTP	55	740	1.2	7	98.8
Hybrid	TCP/HTTP	36	930	0.1	3	99.9

9.2 Results Table — Peak Monitoring Load

PLACEHOLDER — replace with your team's actual work: illustrative example values below — replace with your own measured readings (mean of ≥3 trials) from the simulator.

Topology	Protocol	Avg. Response Time (ms)	Throughput (kbps)	Packet Loss (%)	Jitter (ms)	PDR (%)
Star	UDP	78	640	3.5	11	96.5
Mesh	UDP	61	710	1.4	6	98.6
Bus	UDP	112	520	6.8	18	93.2
Hybrid	UDP	58	730	1.1	5	98.9

9.3 Results Table — Disruption / Congestion

PLACEHOLDER — replace with your team's actual work: illustrative example values below — replace with your own measured readings (mean of ≥3 trials) from the simulator.

Topology	Failure/Load Condition	End-to-End Delay (ms)	Packet Loss (%)	Recovery/Failover Time (s)
Star	Core link saturated	145	9.2	N/A — no redundant path
Mesh	One inter-zone link down	64	1.8	2.1
Bus	Backbone segment fault	N/A — zone isolated	100	N/A — no redundant path
Hybrid	Core link down (reroute)	70	2.0	2.6

PLACEHOLDER — replace with your team's actual work: insert your charts (bar/line graphs comparing topologies per metric) here, exported from the simulator or built from your logged data — required deliverable #6.

10. Analysis, Comparison, Trade-offs and Justification

PLACEHOLDER — replace with your team's actual work: the paragraph and bullets below reason from the illustrative example figures in Section 9 — rewrite them once you substitute your own measured data; the pattern of reasoning (bottleneck → protocol behaviour → trade-off → recommendation) should stay the same.

Across the illustrative example data, the bus topology consistently shows the weakest performance: its shared backbone segment becomes a single collision/failure domain, so under peak monitoring load its response time rises sharply and packet loss climbs, and under the disruption scenario a backbone fault isolates the entire zone with no alternate path. The star topology performs acceptably under normal load but degrades noticeably once the core link is saturated, since every zone funnels through that one device with no redundancy. Mesh and hybrid track each other closely and remain the most stable under both peak load and disruption, because a failed or congested link can be routed around.

- Bottleneck identification: the core/backbone link is the first point of saturation in the star and bus designs; mesh and hybrid distribute load across redundant paths and avoid a single choke point.
- Protocol behaviour: TCP/HTTP traffic degrades more gracefully (retransmission masks some loss but raises response time), while UDP-based monitoring traffic shows loss and jitter more directly since it has no retransmission — this is why peak-load UDP figures are the most sensitive indicator of topology quality.
- Topology trade-offs: bus is cheapest but least resilient; star is simple but has a single point of failure; mesh is most resilient but costliest/most complex to wire and configure; hybrid captures most of mesh's resilience at lower cost than full mesh.
- Final recommendation: hybrid is recommended, since it matches or beats mesh on every metric in the illustrative data while requiring fewer inter-zone links, and clearly outperforms star and bus under both peak-load and disruption scenarios.

11. Broader Considerations and SDG Relevance

This project maps to SDG 3 — Good Health and Well-Being: a reliable hospital network directly supports timely patient monitoring, faster laboratory turnaround, and dependable emergency communication, all of which contribute to improved health outcomes.

- Reliability & patient safety: network downtime in a hospital has direct clinical consequences, unlike in a typical enterprise setting.
- Scalability: the recommended architecture should accommodate additional wards, devices, or telemedicine services without redesign.
- Resource efficiency: bandwidth and device provisioning should avoid unnecessary over-engineering while still meeting peak-load requirements.
- Responsible experimentation: simulated disruption tests should be clearly isolated from any production assumptions and documented transparently.
- Standards compliance: structured cabling (e.g., TIA/EIA-568) and interoperable protocols were used to keep the design vendor-neutral and maintainable.

12. Conclusion, Limitations and Possible Improvements

[Summarize, in 1–2 paragraphs, the final recommended topology and the key evidence supporting it.]

12.1 Limitations

- Simulation results are bounded by the fidelity of the chosen platform and may not fully capture real hospital RF interference, hardware faults, or clinical-grade device behaviour.
- [Add any additional limitations specific to your simulation, e.g., limited trial count, simplified traffic models.]

12.2 Possible Improvements

- Extend testing with a larger device count and additional redundancy paths to validate scalability claims further.
- Incorporate QoS/traffic-prioritization policies for emergency and monitoring traffic in a future iteration.
- [Add further improvements identified by your team.]

13. Individual Contribution of Group Members

Name	Roll No.	Key Contributions
RITESH.Y	192512100	Network topology design (Star & Mesh); device, media and cabling selection; logical/physical topology diagrams
KIRAN.M	192512401	Simulation build in Packet Tracer/GNS3 (Bus & Hybrid); traffic generation (TCP/UDP/HTTP/DNS/FTP/ICMP) across all test scenarios
CHIRANJIBI PRADHAN	192511475	Performance measurement & data collection; results analysis and comparison; report compilation and documentation

PLACEHOLDER — replace with your team's actual work: this is an initial contribution split — adjust it if it doesn't reflect what each of you actually did.

14. References

[List all textbooks, standards documents (e.g., TIA/EIA-568), simulator documentation, datasets, and other sources used, in a consistent citation style, e.g.:]

- [Author/Organization]. (Year). Title of source. Publisher/URL.
- Cisco Packet Tracer / GNS3 / EVE-NG official documentation. [URL].
- [Add remaining references used by your team.]

15. Individual Reflection

Per the assignment instructions, each team member must submit an individual, one-page reflection in their own words. This section is this student's personal reflection space — it must be completed personally, not on the team's behalf, since it is used to assess genuine learning and CO attainment.

The reflection should address:

1. The most challenging aspect of this assignment for you personally.
2. The most useful networking concept you applied or learned.
3. A specific technical problem you encountered and how you solved it.
4. Lessons learned from using the simulation tool(s).
5. How this assignment connects to SDG 3 (Good Health and Well-Being).
6. One engineering decision you would change if you redid this assignment, and why.
7. How this assignment improved your understanding of computer networks and contributed to the mapped Course Outcomes (CO3, CO4, CO5).

CHIRANJIBI PRADHAN — 192511475

Individual Reflection (300–500 words):
Working on the Smart Healthcare Network assignment was a valuable learning experience because it allowed me to apply computer networking concepts to a realistic healthcare environment. My main responsibility in the project was performance measurement and data collection, analysing and comparing the results, and compiling the report. This role helped me understand that designing a network is not only about connecting devices but also about evaluating whether the network can provide reliable and efficient communication under different conditions.

The most challenging aspect for me was analysing network performance across different topologies and traffic scenarios. Comparing Star, Mesh, Bus, and Hybrid topologies required me to consider several performance indicators, including response time, throughput, packet loss, jitter, packet delivery ratio, and end-to-end delay. I learned that a topology that appears simple and inexpensive may not necessarily be suitable for a critical environment such as a hospital. The Hybrid topology was particularly interesting because it combines the manageability of a Star topology with the redundancy and fault tolerance of a Mesh-based core.

The most useful networking concept I applied was understanding the difference between TCP and UDP and how their behaviour changes under network load. TCP provides reliable communication through retransmission, making it appropriate for applications such as EMR and file transfer. UDP is more suitable for time-sensitive monitoring traffic, but packet loss and jitter become more visible because it does not provide retransmission. This helped me understand why network performance must be evaluated according to the requirements of the application.

One technical challenge was interpreting performance data during peak-load and disruption scenarios. I addressed this by comparing the metrics systematically across the different topologies and identifying bottlenecks such as saturated core or backbone links. The simulation also taught me that repeated trials and structured measurement are important for obtaining more reliable observations rather than depending on a single result.

This assignment also helped me understand the connection between networking and **SDG 3 — Good Health and Well-Being**. In a hospital, reliable communication can support patient monitoring, laboratory information, medical records, and emergency communication. Therefore, network reliability can indirectly contribute to better healthcare services and patient safety.

If I repeated the project, I would improve the design by giving greater attention to QoS and traffic prioritisation for emergency and patient-monitoring traffic. Overall, this assignment improved my practical understanding of network topology, protocols, performance analysis, and fault tolerance. It also strengthened my ability to analyse technical results and make engineering decisions based on evidence, contributing to the mapped **CO3, CO4, and CO5**.

16. Submission Checklist

#	Deliverable	Status
1	Technical Report (this document)	In progress
2	Network Topology Diagrams (logical + physical, all 4 designs)	[ ]
3	Simulation/Emulation Project Files	[ ]
4	Experiment & Test Plan	Included — Section 6
5	Protocol Performance Evidence (captures/screenshots)	[ ]
6	Performance Measurement Results (raw + processed)	[ ]
7	Engineering Analysis & Design Justification	Included — Section 10

8	Simulation Configuration & Validation Evidence	[ ]
9	Individual Reflection (per student)	Template included — Section 15
10	Presentation / Demonstration	[ ]
11	Pseudocode	Included — Section 7
12	GitHub Upload (full project + source/config files)	[ ]